Forecasting - Graded Exercise

Mirjeta Misini

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Packages

```
library(car)
```

```
## Loading required package: carData
```

Question 2 :

Introduction to simulations

Simulating paths of forecasts can be useful in many cases, for instance when (Pittavino, 2019) :

- There is no explicit formula for the distribution of Yt
- The situations is computationally complicated

Methodolgy :

- Simulation of a random sample of independent errors of distribution e ~ N(0, σ^2)
- Apply the adequate state space model to build random paths
- Extract the characteristics needed, for example the mean and standard deviation for prediction intervalls quantiles.

Bootstrap

Sometimes, the normal simulation cannot be used. Instead re-sampling the fitted errors of our sample, with replacement, can help. This is called a boostrap method.

It follows the same steps as the normal simulation, except that the errors simulated are chosen from the set with equal probability *1/t*.

Prediction intervalls

To get these intervalls of forecasts, the [R] simulated paths *y(t+1)*[1],…, *y(t+1)*[R] must be ordered from the smallest to the largest. Then, the computation of the quantiles at *α/2* and *1-α/2* gives back the intervall limits.

An application of the prediction intervalls with bootstrap and normal simulations is provided on a dataset at the end of this paper.

Question 3 :

Introduction to Inventory Management

In order to continuously satisfy the demand, its amount should be predicted wisely. However it's often not a constant measure and the uncertainty behind it should be taken into account when forecasting. Therefore, it has a variance, and "in classical models, demand per period of time *t* is assumed to be normally distributed" (Pittavino, 2019) :

$$D_t \sim N(D, \sigma_D^2).$$

with a mean **D** and a standard deviation **σD**.

This uncertainty can have many causes, from natural variations, to unexpected events.

The most common factors of the demand uncertainty are the following (Mithu, Roy, 2013) :

- **Quality/Price of the product of competitors** : The demand can suddenly change unexpectedly if the competitors have higher/lower quality products or if they change their prices.
- **Affordability of the customers** : If customers can no longer economically afford to buy the products, or depending on the type of product, if they have a large increase in their income, the producer/distributor may not be able to instantly anticipate the variation.
- **Up-coming product** : New technologies or trends are often hard to predict very early, and it can modify the demand rapidely.
- **Availability of the product in the market** : Retailers/Distributors may decide to sell more of some products and suddenly stop offering others. This directly affects the demand in many ways. Customers may start engaging with some other brands/products, or they may completely change their buying habits, like changing the place they would usually go to buy for.
- **Variation of lead time** : Delays in delivering the products force the demand to unexpectedly vary.
- **Seasonal effect** : Seasonal fluctuations.

For all these factors, the following concepts are required to be studied and well-predicted. Without aknowledging these terms, the demand would not be satisfied except in the hypothetic case of a deterministic demand, meaning that the demand would never face uncertainty. This assumption is evidently not often met in reality.

```
The concept of safety stock (SS)
```

The safety stock helps to cope with these uncertainties and to avoid facing out-of-stock situations. However this safety stock has a price. If it's really large the higher cost is due to (Pittavino, 2019) :

- Storing
- The loss of unsold products
- The degradation of quality, etc.

If on the opposite this stock is low, loosing customers and reputation can have large negative impacts on the firm. The forecast of this safety stock should then be precise.

To find the amount of this stock, the firm needs to get the **Reorder Point (ROP)** given a certain **Service Level (SL)** it wants to provide. The difference between the **expected demand during lead time (E[DL])** and the **ROP** gives back the **safety stock (SS)**. These notions in bold are all explained below.

```
The concept of lead time (L)
```

The lead time is the amount of time needed for the ordered products to arrive. It's expressed in **L** numbers of periods (expressed on the same unit of time as the demand).

This is an important component in inventory management for many reasons (Pittavino, 2019) :

- The demand during this time, called **DL** must still be satisfied.
- There may be unexpected delays that will impact the offer of the products.
- It helps to determine the reoder point, "the stock at which an order of replenishment is set".

In order to compute the **DL**, the demand that must be satisfied during this lead time, we need to sum the independent demands from period *t* to *t+L*. Under the normal distribution assumption, it's summing the mean of the demand L-times, so it's expressed by the formula :

$$D_L \sim N(L \times D, L\sigma_D^2)$$

with a mean **L x D** and a standard deviation √**L x σD**.

- Example : if the mean of **D** is 100 units **per week**, the standard deviation of **D** is 30 units, the lead time is about 3 **weeks**, then the mean of **DL** is 300 units (100 x 3) and its standard deviation is 52 (√3 x 30). The **DL** is expressed as :

  $$D_L \sim N(300, 52^2)$$

  The concept of service level (SL)

The service level **SL** is the probability of meeting the expectations of the demand during the lead time, with the stock **S** available :

$$P(D_L \leq S) = SL.$$

```
The concept of reorder point (ROP)
```

This is the moment when the firm should make an order. It depends on the forecast of the demand during lead time and the service level the firm wants to offer.

We can compute it by fixing the service level **SL**, or by summing the safety stock **SS** and the expected demand during lead time **E(DL)**.

When the firm makes an order at each time the ROP quantity in stock is reached, it follows a **continuous replenishment policy**.

If the firms decides to reorder at regular intervalls of time, it's called a **periodic replenishment policy**. In this case, the ROP changes, because the demand during lead time **D(L)** becomes **D(T+L)**, where T is the regular intervall of control.

Application to the dataset : "elecquip"

This dataset is about orders for electrical equipments in the Eurozone area (16 countries)

```
library(fpp)
```

```
## Loading required package: forecast
```

```
## Registered S3 method overwritten by 'xts':
##   method     from
##   as.zoo.xts zoo
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
## Registered S3 methods overwritten by 'forecast':
##   method             from
##   fitted.fracdiff    fracdiff
##   residuals.fracdiff fracdiff
```

```
## Loading required package: fma
```

```
## Loading required package: expsmooth
```

```
## Loading required package: lmtest
```

```
## Loading required package: zoo
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
## Loading required package: tseries
```

```
elec<- fpp::elecequip
frequency(elec)
```

```
## [1] 12
```

```
start(elec)
```

```
## [1] 1996    1
```

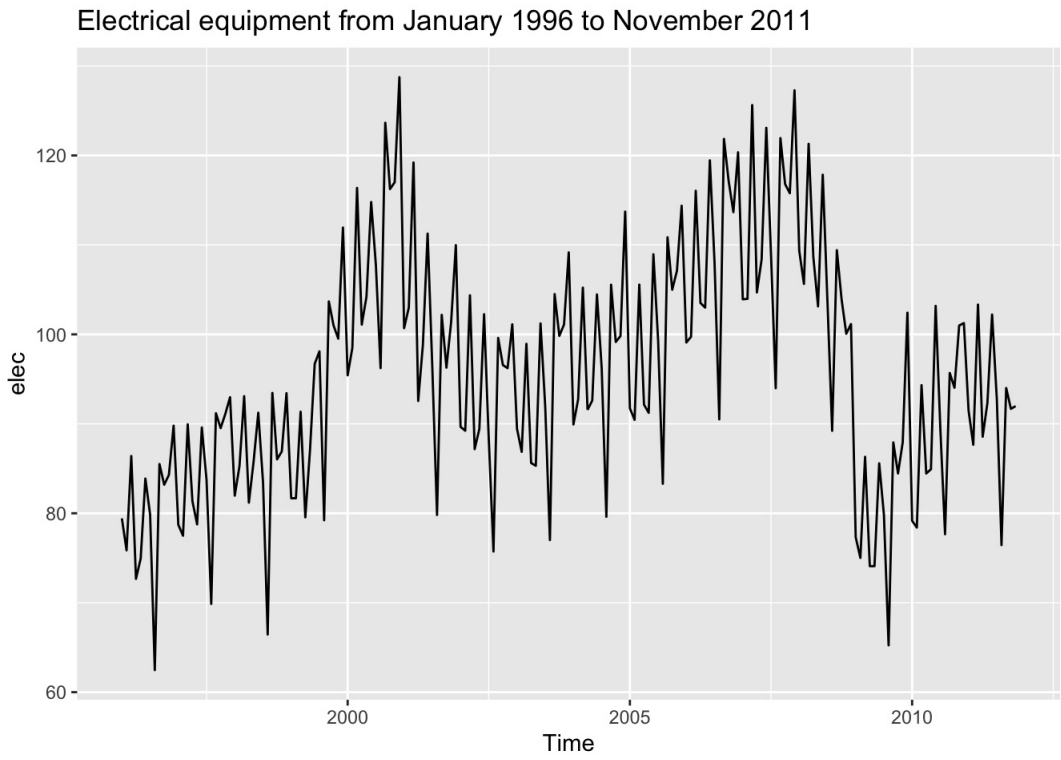
```
end(elec)
```

```
## [1] 2011   11
```

```
str(elec)
```

```
##  Time-Series [1:191] from 1996 to 2012: 79.4 75.9 86.4 72.7 74.9 ...
```

```
autoplot(elec, main="Electrical equipment from January 1996 to November 2011")
```

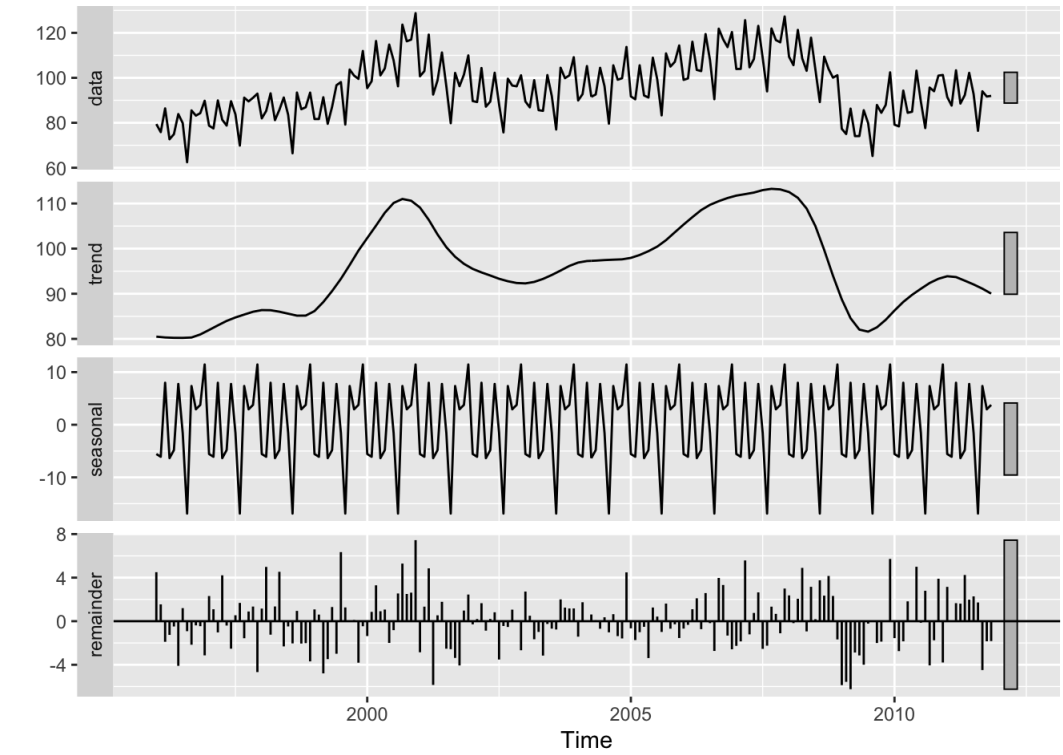


Decomposition of the time-series

```
mstl_elec<-mstl(elec)
stl_elec<-stl(elec, s.window = "periodic")

# The first 5 points (x) and their decomposition
cbind(as.data.frame(head(elec,5)),as.data.frame(head(stl_elec$time.series,5)))
```

```
##       x  seasonal    trend  remainder
## 1 79.43 -5.563267 80.49243  4.5008362
## 2 75.86 -6.088288 80.40279  1.5454971
## 3 86.40  7.977940 80.31315 -1.8910917
## 4 72.67 -6.347420 80.27340 -1.2559820
## 5 74.93 -4.818401 80.23365 -0.4852496
```

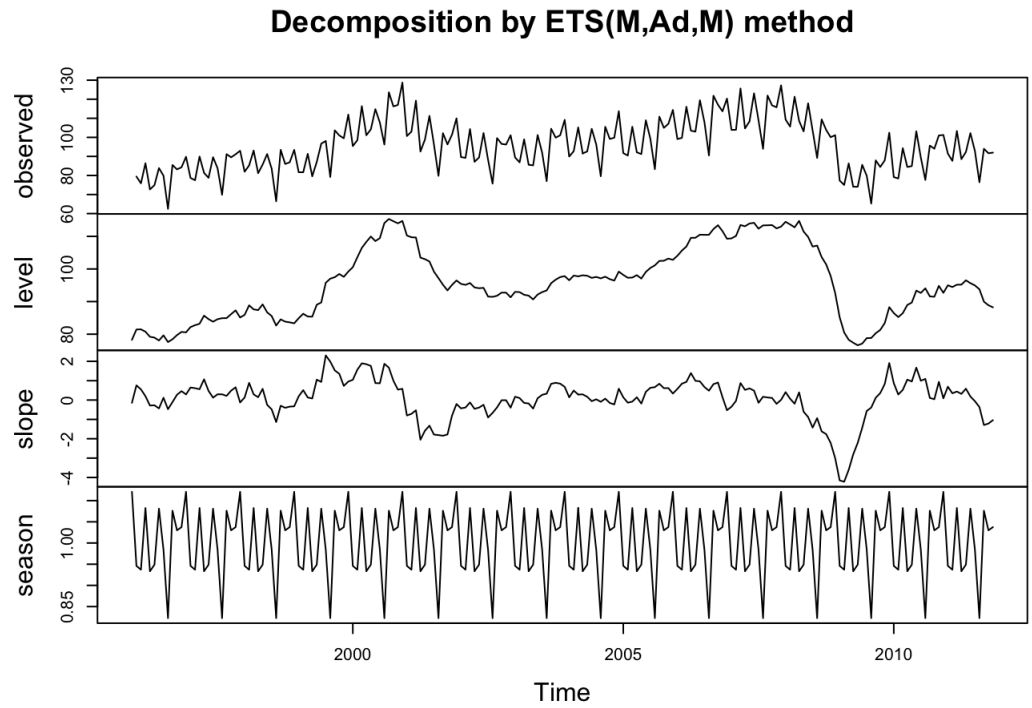
```
autoplot(stl_elec)
```



The choice of a model

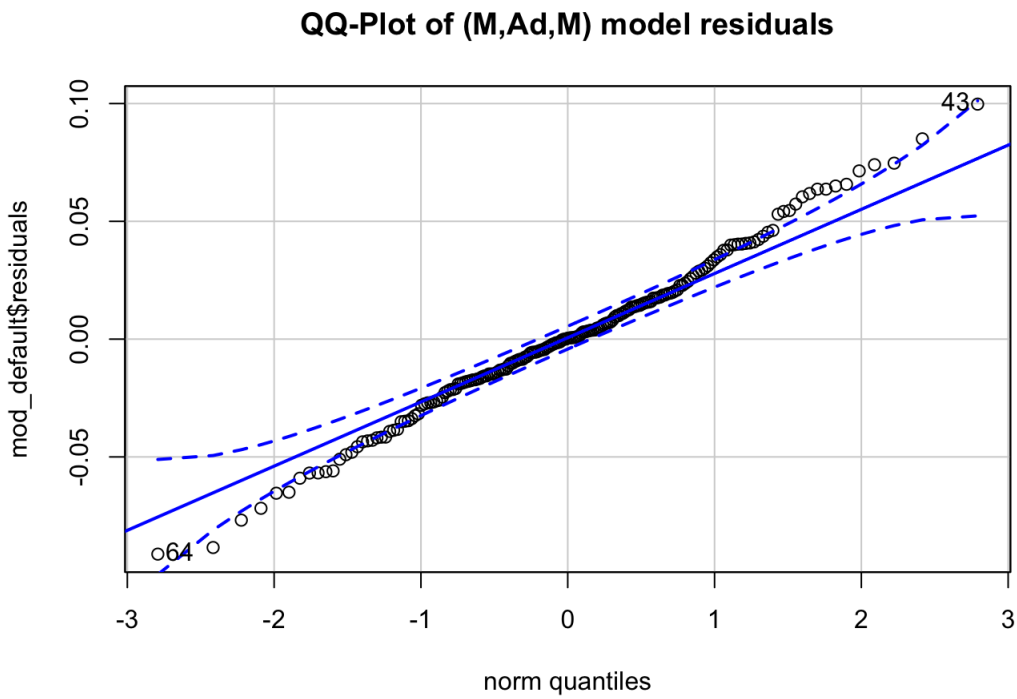
The default model given by R :

```
set.seed(1)
mod_default<-ets(elec, model="ZZZ")
plot(mod_default)
```



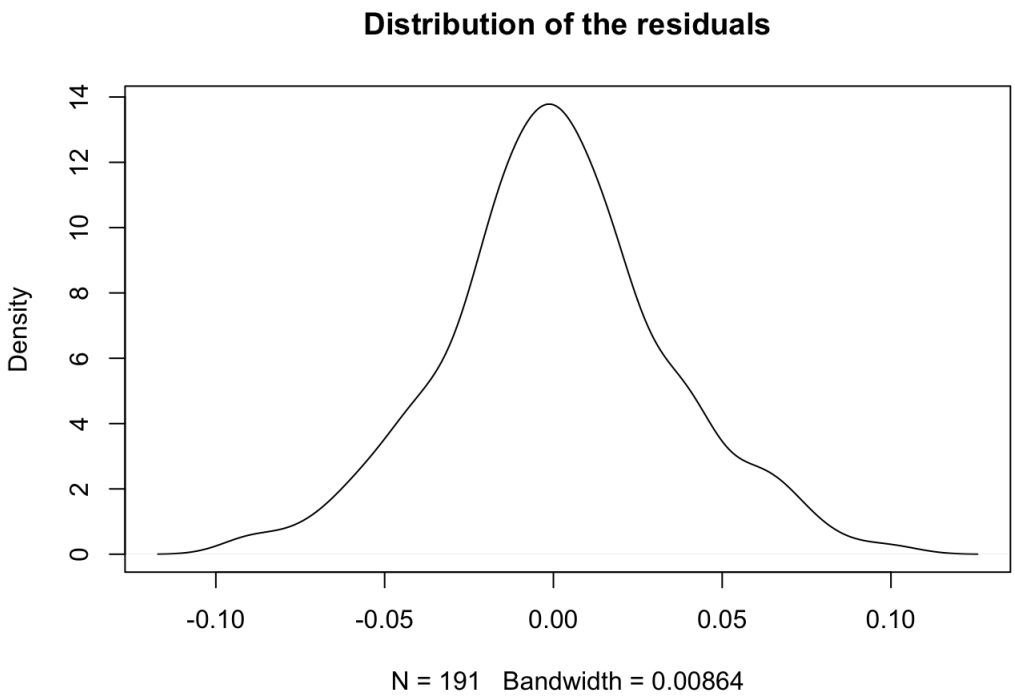
The model chosen by R is the one with multiplicative errors, additive damped trend, and multiplicative seasonality. Let's check the normality and the centered mean of the residuals.

```
qqPlot(mod_default$residuals, main = "QQ-Plot of (M,Ad,M) model residuals")
```




```
## [1] 43 64
```

```
plot(density(mod_default$residuals), main = "Distribution of the residuals")
```



The shape looks symmetric and centered around zero.

The tests

```
# T-test
t.test(mod_default$residuals)
```

```
##
##  One Sample t-test
##
## data:  mod_default$residuals
## t = 0.35204, df = 190, p-value = 0.7252
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  -0.003898725  0.005592684
## sample estimates:
##    mean of x
## 0.0008469792
```

```
# Shapiro-Wilk test
shapiro.test(mod_default$residuals)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  mod_default$residuals
## W = 0.9934, p-value = 0.5508
```

The t-test couldn't reject the null hypothesis of zero mean, and the shapiro-wilk test couldn't reject as well the null hypothesis of normality distribution.

Based on the *Midterm* paper of my colleague *Alexandre Tchittarath* where he explored multiple types of models with a cross-validation method, this model can be chosen as it gives back the lowest RMSE :

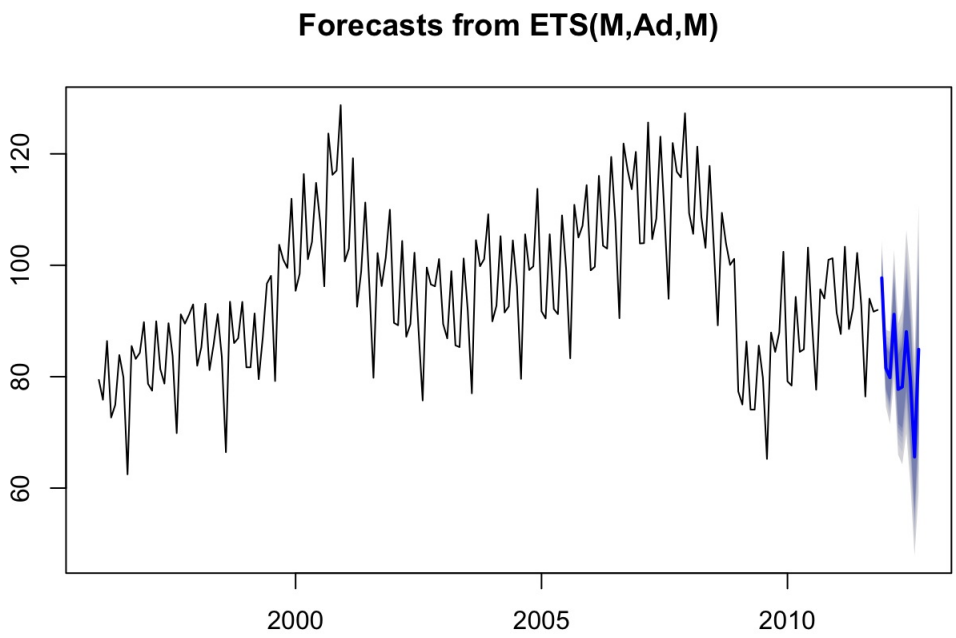
```
##    Optimal      MAM     MAdM     MAdA   AAdA      AAA
## 1 9.045695 9.984434 9.878761 8.912806 8.9618 10.29903
```

Under "Optimal". (Tchittarath, 2019)

Let's forecast now for 10 step ahead.

Forecast with the function forecast() :

```
set.seed(1)
forecast_10<-forecast(mod_default, h=10, level = seq(0.7,0.95,0.05))
plot(forecast_10)
```



Forecasting using bootstrap method with assumption of residuals normality :

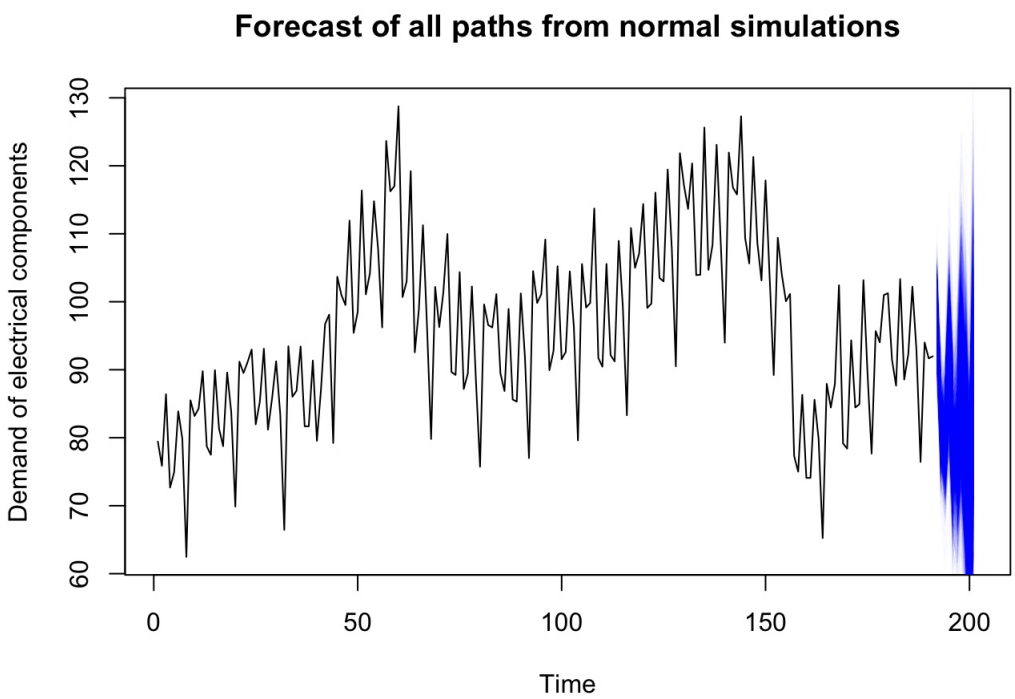
```
## Parameters/Coeff
alpha<-mod_default$par[1]
beta<-mod_default$par[2]
gamma<-mod_default$par[3]
phi<-mod_default$par[4]
sigma<-sqrt(mod_default$sigma2)

## Seasonal index
states<-data.frame(mod_default$states)
seasonalities<-states[,-c(1,2)]
S_index<-matrix(nr=1, nc=12)
i_seas<-c(12:1)
for (i in seq(1:12)){
  S_index[i]<-seasonalities[192,i_seas[i]]
}

seasonalities[192,]
```

```
##           s1       s2       s3        s4        s5       s6        s7
## 192 1.037496 1.029914 1.076202 0.8227363 0.9841509 1.080999 0.9488557
##             s8      s9       s10       s11      s12
## 192 0.9333261 1.08255 0.9370097 0.9460197 1.120757
```

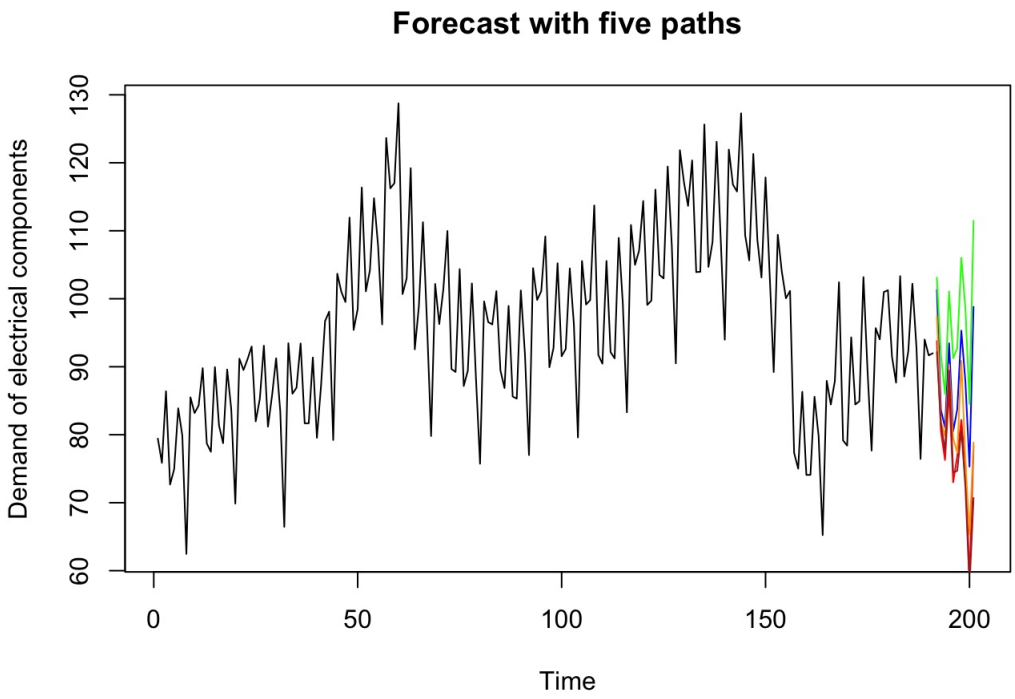
```
S_index
```

```
##          [,1]      [,2]      [,3]    [,4]      [,5]      [,6]     [,7]
## [1,] 1.120757 0.9460197 0.9370097 1.08255 0.9333261 0.9488557 1.080999
##           [,8]      [,9]    [,10]    [,11]    [,12]
## [1,] 0.9841509 0.8227363 1.076202 1.029914 1.037496
```

```r
## Creation of the 5000 paths matrices
set.seed(1)
eps<-matrix(rnorm(10*5000, sd = sigma), nr=10, nc=5000)
Y<-matrix(nr=10,nc=5000)
L<-matrix(nr=10, nc=5000)
B<-matrix(nr=10, nc=5000)
S<-matrix(nr=10, nc=5000)

## State at t0
L0<-states[192,1]
B0<-states[192,2]


## States at t+1 (December 2011)
Y[1,]<-(L0+(phi*B0))*S_index[[1]]*(1+eps[1,])
L[1,]<-(L0+(phi*B0))*(1+(alpha*eps[1,]))
B[1,]<-(phi*B0)+(beta*(L0+(phi*B0))*eps[1,])
S[1,]<-S_index[[1]]*(1+(gamma*eps[1,]))

## States for next months (t+2 to t+10)
for (i in 2:10){
  Y[i,]<-(L[i-1,]+(phi*B[i-1,]))*S_index[[i]]*(1+eps[i,])
  L[i,]<-(L[i-1,]+(phi*B[i-1,]))*(1+(alpha*eps[i,]))
  B[i,]<-(phi*B[i-1,])+(beta*(L[i-1,]+(phi*B[i-1,]))*eps[i,])
  S[i,]<-S_index[[i]]*(1+(gamma*eps[i,]))
}

plot(as.numeric(elec), type="l", xlim=c(1,202), ylab="Demand of electrical components", xlab="Time", main="F
orecast of all paths from normal simulations")

for(i in seq(5000)){

points(Y[,i]~c(192:201), type = "l", lwd = 0.01, col = "blue")

}
```

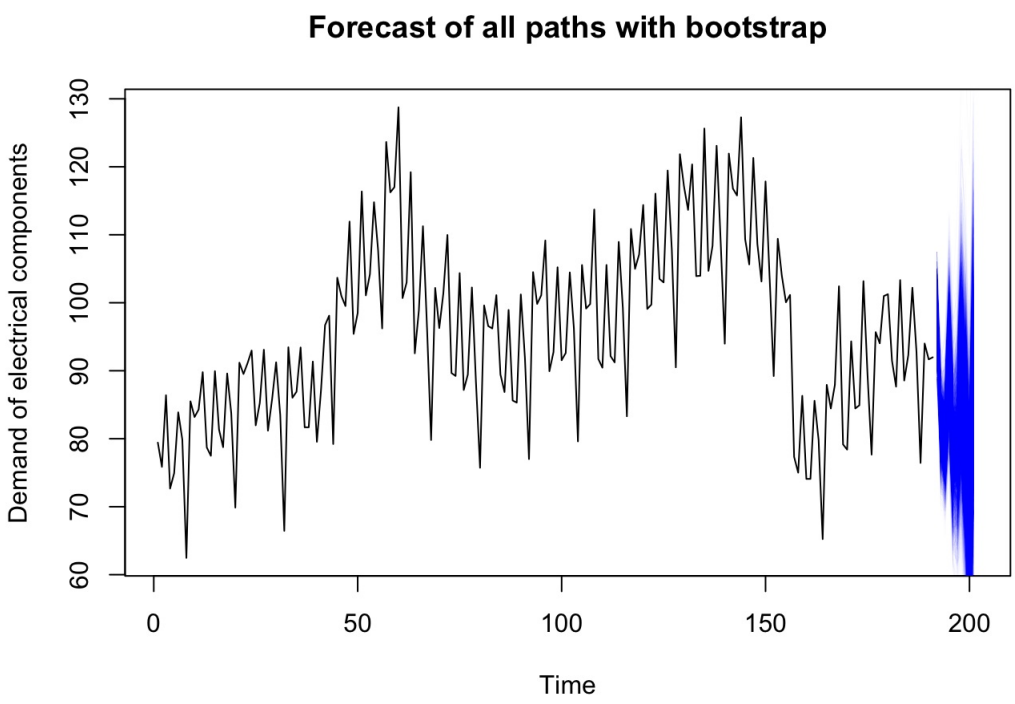


```
plot(as.numeric(elec), type="l", xlim=c(1,202), ylab="Demand of electrical components", xlab="Time", main="F
orecast with five paths")
j=1
for(i in c(1000,2000,3000,4000,5000)){
  col=c("blue","red","green","orange","brown")
  points(Y[,i]~c(192:201), type = "l", lwd = 1, col = col[j])
  j=j+1
}
```



Forecasting using a bootstrap method by sampling with replacement the fitted model's residuals

```
set.seed(1)
eps2 = matrix(sample(size = 10*5000, x = mod_default$residuals, replace = TRUE), nr=10, nc=5000)
Y1<-matrix(nr=10,nc=5000)
L1<-matrix(nr=10, nc=5000)
B1<-matrix(nr=10, nc=5000)
S1<-matrix(nr=10, nc=5000)

## State at t0
L0<-states[192,1]
B0<-states[192,2]


## States at t+1 (December 2011)
Y1[1,]<-(L0+(phi*B0))*S_index[[1]]*(1+eps2[1,])
L1[1,]<-(L0+(phi*B0))*(1+(alpha*eps2[1,]))
B1[1,]<-(phi*B0)+(beta*(L0+(phi*B0))*eps2[1,])
S1[1,]<-S_index[[1]]*(1+(gamma*eps2[1,]))

## States for next months (t+2 to t+10)
for (i in 2:10){
  Y1[i,]<-(L1[i-1,]+(phi*B1[i-1,]))*S_index[[i]]*(1+eps2[i,])
  L1[i,]<-(L1[i-1,]+(phi*B1[i-1,]))*(1+(alpha*eps2[i,]))
  B1[i,]<-(phi*B1[i-1,])+(beta*(L1[i-1,]+(phi*B1[i-1,]))*eps2[i,])
  S1[i,]<-S_index[[i]]*(1+(gamma*eps2[i,]))
}

plot(as.numeric(elec), type="l", xlim=c(1,202), ylab="Demand of electrical components", xlab="Time", main="F
orecast of all paths with bootstrap")

for(i in seq(5000)){

points(Y1[,i]~c(192:201), type = "l", lwd = 0.01, col = "blue")

}
```



The paths look very similar between the normal simulation and the bootstrap method.

The prediction intervalls

```r
## From the forecast function

PI_forecast<-matrix(nr=10,nc=12)
j=seq(1,11,2)
for (i in seq(1,6)){
  PI_forecast[,j[i]]<-forecast_10$lower[,i]
  PI_forecast[,j[i]+1]<-forecast_10$upper[,i]
}
colnames(PI_forecast)<-c("15%", "85%", "12.5%", "87.5%","10%", "90%","7.5%", "92.5%","5%", "95%","2.5%", "97.5%")


## From the normal simulation

PI_norm<-matrix(nr=10, nc=12)
probs<-c(0.15, 0.85, 0.125, 0.875,0.1,0.9,0.075, 0.925 ,0.05, 0.95,0.025, 0.975)
for (i in seq(1,12)){
  quant<-apply(Y, 1, quantile, probs=probs[i])
  PI_norm[,i]<-quant
}
colnames(PI_norm)<-c("15%", "85%", "12.5%", "87.5%","10%", "90%","7.5%", "92.5%","5%", "95%","2.5%", "97.5%")


## From the bootstrap

PI_bootstrap<-matrix(nr=10, nc=12)
for (i in seq(1,12)){
  quant<-apply(Y1, 1, quantile, probs=probs[i])
  PI_bootstrap[,i]<-quant
}
colnames(PI_bootstrap)<-c("15%", "85%", "12.5%", "87.5%","10%", "90%","7.5%", "92.5%","5%", "95%","2.5%", "97.5%")

PI_forecast[,1:6]
```

```
##            15%       85%    12.5%     87.5%      10%       90%
##  [1,] 94.21054 101.25166 93.82359 101.63861 93.37792 102.08428
##  [2,] 77.89839  85.17140 77.49869  85.57109 77.03835  86.03144
##  [3,] 75.44162  84.22024 74.95918  84.70267 74.40354  85.25832
##  [4,] 85.08968  97.27898 84.41981  97.94886 83.64829  98.72038
##  [5,] 71.52041  83.94821 70.83743  84.63119 70.05081  85.41780
##  [6,] 70.79775  85.50903 69.98928  86.31750 69.05813  87.24866
##  [7,] 78.44248  97.68718 77.38487  98.74479 76.16678  99.96289
##  [8,] 69.37353  89.24718 68.28136  90.33935 67.02345  91.59726
##  [9,] 56.27338  74.91967 55.24866  75.94439 54.06844  77.12461
## [10,] 71.34252  98.46495 69.85198  99.95548 68.13527 101.67219
```

```r
PI_norm[,1:6]
```

```
##            15%       85%    12.5%     87.5%      10%       90%
##  [1,] 94.12526 101.23395 93.70726 101.60328 93.24373 102.08633
##  [2,] 77.90521  85.22920 77.43251  85.63228 76.93020  86.07215
##  [3,] 75.46017  84.28178 74.99679  84.86103 74.45435  85.29877
##  [4,] 85.34459  97.45604 84.66468  98.09067 83.85011  98.88167
##  [5,] 71.63248  84.12099 71.03667  84.83731 70.24170  85.60904
##  [6,] 71.00034  85.64524 70.21842  86.43648 69.47029  87.44349
##  [7,] 78.56120  97.88034 77.64256  98.97699 76.41192 100.22395
##  [8,] 69.52235  89.46914 68.41729  90.67677 67.22937  91.86353
##  [9,] 56.42586  75.19424 55.54396  76.18829 54.44479  77.54467
## [10,] 71.65496  98.60819 70.29951 100.16571 68.83876 102.03143
```

```r
PI_bootstrap[,1:6]
```

```
##            15%       85%    12.5%     87.5%      10%       90%
##  [1,] 94.45937 101.26572 93.97552 101.70034 93.66305 101.89349
##  [2,] 78.31626  84.99943 77.91122  85.48644 77.51100  86.00347
##  [3,] 75.81331  84.11972 75.33595  84.65432 74.78122  85.20032
##  [4,] 85.87243  97.35709 85.23348  97.91412 84.46127  98.71251
##  [5,] 72.32174  84.04309 71.66953  84.68825 70.89878  85.58543
##  [6,] 71.59944  85.61083 70.89793  86.46971 70.12600  87.40451
##  [7,] 79.61759  97.86019 78.66803  99.06420 77.45745 100.52449
##  [8,] 70.65627  89.36860 69.79803  90.64062 68.61255  92.09452
##  [9,] 57.45521  75.24626 56.48611  76.34778 55.41411  77.74697
## [10,] 72.89845  98.85475 71.47586 100.37371 70.02702 102.24812
```

```
PI_forecast[,7:12]
```

```
##           7.5%     92.5%       5%       95%     2.5%     97.5%
##  [1,] 92.84129 102.62091 92.14385 103.31835 91.07348 104.38872
##  [2,] 76.48405  86.58574 75.76364  87.30615 74.65802  88.41176
##  [3,] 73.73449  85.92736 72.86495  86.79691 71.53045  88.13140
##  [4,] 82.71930  99.64937 81.51192 100.85674 79.65894 102.70972
##  [5,] 69.10365  86.36497 67.87265  87.59597 65.98342  89.48520
##  [6,] 67.93693  88.36985 66.47975  89.82704 64.24339  92.06340
##  [7,] 74.70008 101.42959 72.79384 103.33582 69.86833 106.26134
##  [8,] 65.50882  93.11189 63.54029  95.08042 60.51916  98.10155
##  [9,] 52.64735  78.54570 50.80039  80.39266 47.96584  83.22721
## [10,] 66.06818 103.73928 63.38164 106.42582 59.25858 110.54888
```

```
PI_norm[,7:12]
```

```
##           7.5%     92.5%       5%       95%     2.5%     97.5%
##  [1,] 92.65895 102.67197 91.94925 103.36460 90.91444 104.27217
##  [2,] 76.32826  86.68595 75.69374  87.43865 74.56676  88.61600
##  [3,] 73.74092  86.05556 72.96454  86.93210 71.57517  88.36800
##  [4,] 82.93134  99.88946 81.83980 101.14892 80.04681 103.06988
##  [5,] 69.29969  86.55043 68.26731  87.93872 66.50502  89.96531
##  [6,] 68.48935  88.76693 67.15705  90.44392 65.32655  92.70878
##  [7,] 75.13716 101.97781 73.52222 104.28956 71.25482 108.13238
##  [8,] 66.12240  93.51974 64.50177  95.72604 62.07090  99.45114
##  [9,] 53.20483  78.92753 51.76342  80.99469 49.48880  84.34247
## [10,] 67.06434 104.46498 64.72318 107.30072 62.04832 112.14246
```

```
PI_bootstrap[,7:12]
```

```
##           7.5%     92.5%       5%       95%     2.5%     97.5%
##  [1,] 93.06427 103.05639 92.20562 103.98062 91.36436 104.73920
##  [2,] 76.94941  86.59297 76.31158  87.32177 75.21637  88.53981
##  [3,] 74.15227  85.95427 73.42135  86.76392 72.27039  87.99664
##  [4,] 83.57992  99.55343 82.57956 100.75053 80.70584 102.72604
##  [5,] 70.05597  86.48965 68.77791  87.72482 67.04718  89.49621
##  [6,] 69.21007  88.77886 67.64106  90.28150 65.77379  92.50280
##  [7,] 76.14147 102.26199 74.69261 104.29388 72.15844 107.09930
##  [8,] 67.37252  93.72329 65.58885  95.94638 62.94745  99.06701
##  [9,] 54.31379  79.39653 52.70312  81.47685 50.36810  84.11271
## [10,] 68.67423 104.39944 66.37685 107.39930 62.94651 112.26951
```

The prediction intervalls are quite similar between the three methods.

Another remark can be made about the length of the intervalls in general. The further periods we predict, the larger are the intervalls because previous errors are added at each period. Indeed, this is due to the uncertainty that increases with the horizon of prediction.

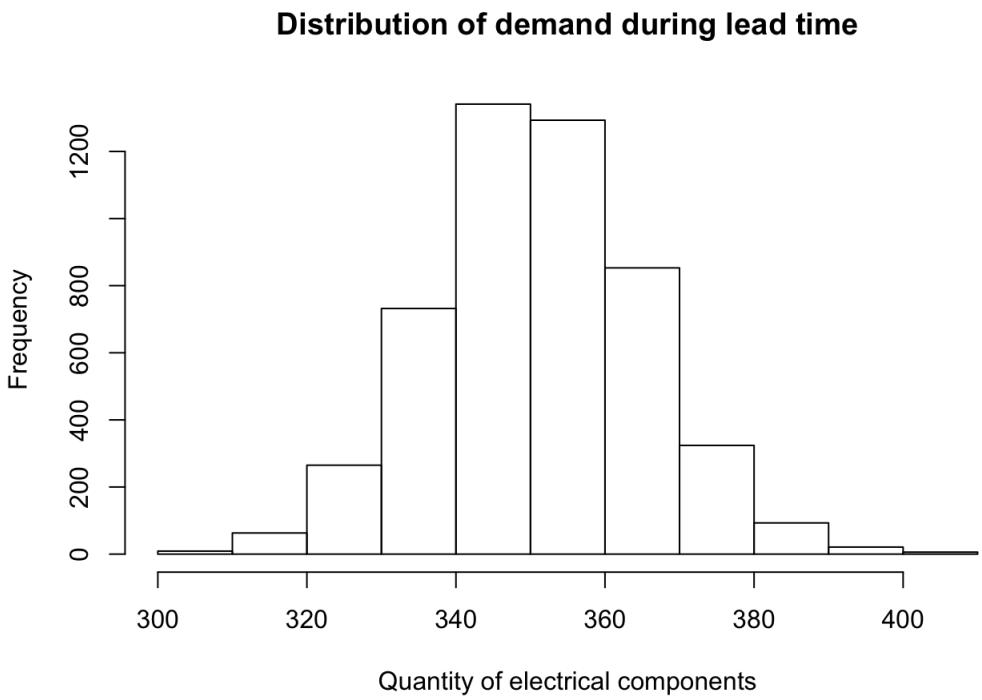
Application of notions from Inventory Management chapter

The ROP

Continuous replenishment policy

Let's assume from now on that the lead time is about 4 months. We'll compute the ROP for different service levels.

```
D_L <- apply(Y1[1:4,], 2, sum)
hist(D_L, main="Distribution of demand during lead time", xlab="Quantity of electrical components")
```



The demand during lead time is mainly between 340 and 360 units.

Let's see the quantity of stock that will push the firm to make an order given different quality of service they want to offer.

```
ROP1 <- matrix(nr=1,nc=6)
colnames(ROP1)<-c("70%","75%","80%","85%","90%","95%")
SL<-seq(0.7,0.95, 0.05)
for (i in c(1:6)){
  ROP1[1,i]<- quantile(D_L, prob=SL[i])
}
ROP1
```

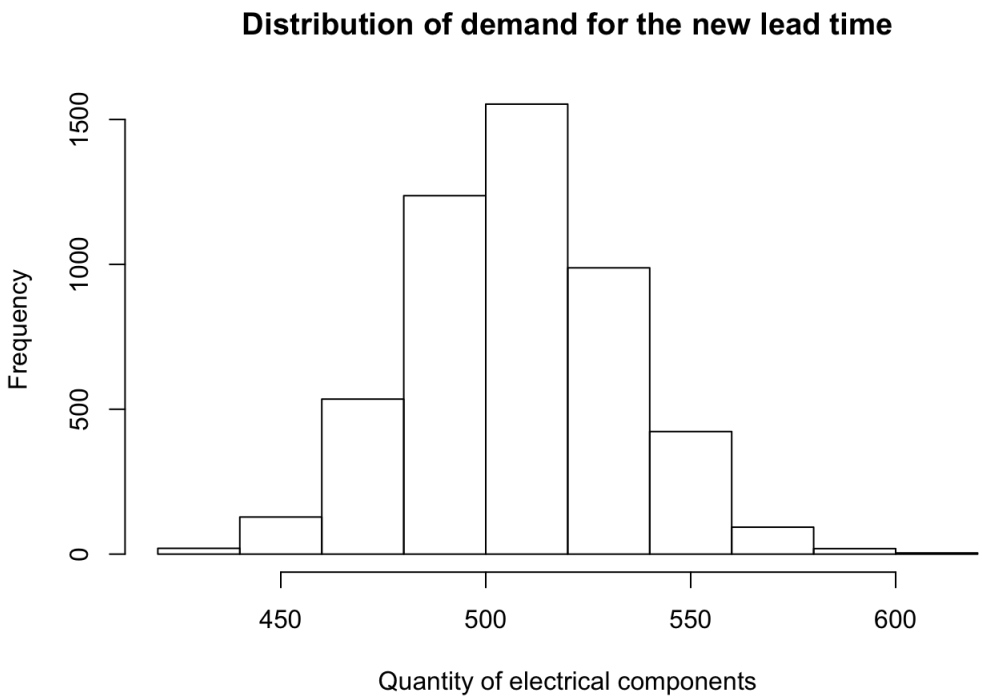
```
##           70%      75%      80%      85%     90%      95%
## [1,] 358.1057 360.4706 362.6677 365.4546 369.029 374.3373
```

In order to assure a very good service level, the amount in stock should be of 375 units otherwise the firm should make a new order.

Periodic replenishment policy

Now, if the stock is periodically reviewed every two months, the ROP changes because it must assure the demand during 6 months.

```
D_L2 <- apply(Y1[1:6,], 2, sum)
hist(D_L2, main="Distribution of demand for the new lead time", xlab="Quantity of electrical components")
```




```
ROP2<-ROP1
for (i in c(1:6)){
  ROP2[1,i]<- quantile(D_L2, prob=SL[i])
}
ROP2
```

```
##           70%      75%      80%      85%      90%      95%
## [1,] 520.4019 524.7078 528.9083 533.6539 541.3474 550.4859
```

The ROP increases linearly as the demand during lead time grows. For a service level of 95% the stock should be at least of 551 units, otherwise the firm will make a new order.

The Newsvendor Model

Let's assume now that we have a situation where the components sold by the firm are perishable. In this case, we use a model called "The Newsvendor Model", as selling newspapers in the streets are the perfect way to explain this model, given that the products cannot be sold after a certain period of time.

In this case, we have to take into account new parameters, the selling price p and the cost c of the product sold.

The main goal of the seller is to determine the quantity q in order to maximize his profits under the demand uncertainty.

A possible way to solve this is to simulate D and estimate the profit for a sequence of q, and look for its maximum.

Another option is to look for the **critical fractile**.

The intuition behind is to add a quantity only if the marginal profit is not zero. However the quantity may not be sold due to the uncertainty of the demand, so the firm should consider the **probability of selling** that quantity, called **PS**.

The expected profit from that quantity is then : $\Delta(q) = p \times PS - c.$

The profit (*delta q*) should be positive so, PS > c/p.

Moreover, the probability of selling that quantity is related to the fact that the demand should exceed the quantity we have plus one, that is when D > q+1.

Thus,

$$PS = P(D \geq q+1) = 1 - P(D \leq q) = 1 - F(q).$$

and

$$1 - F(q) \geq c/p \iff F(q) \leq \frac{p-c}{p}.$$

To find the optimal **(q)**, the firm must increase **q** until this condition above is not satisfied. In other words, it should stop when **F(q)** is bigger

than the righten part of the equation. The optimal solution is therefore found when :

$$q^* = F^{-1}\left(\frac{p-c}{p}\right).$$

This **(q)** is called the **critical fractile**.

Application of the Newsvendor model to the dataset.

Let's suppose the electrical products are perishable **and last 4 months only**. Their cost for the firm is 30 and they're sold for 120. The lead time is about 4 months. What is the quantity that should be ordered ?

Q when using the forecast() function :

```
set.seed(1)
forecast_10<-forecast(ets(elec), h=10)
forecast_10$mean
```

```
##           Jan      Feb      Mar      Apr      May      Jun      Jul
## 2011
## 2012 81.53489 79.83093 91.18433 77.73431 78.15339 88.06483 79.31036
##           Aug      Sep Oct Nov      Dec
## 2011                           97.73110
## 2012 65.59652 84.90373
```

```
DL_mean<-(sum(forecast_10$mean[1:4]))
DL_mean
```

```
## [1] 350.2813
```

This is the expected demand during lead time, computed with the function *forecast()*.

```
## Ratio : (p-c)/p
ratio<-((120-30)/120)
DL_sigma<- sqrt(3)*sigma
DL_sigma
```

```
## [1] 0.06020029
```

```
critic_fr1 <- qnorm(p = ratio, mean = DL_mean, sd = DL_sigma)
ceiling(critic_fr1)
```

```
## [1] 351
```

The quantity that should be in stock at the end of November 2011 is 351 units.

Q when using bootstrap :

```
## D_L is the matrix of bootstrap-ed demands during lead time
mean_D_L<-mean(D_L)
ceiling(mean_D_L)
```

```
## [1] 351
```

```
sigma_D_L<-sqrt((sum((D_L-mean_D_L)^2))/4999)
sigma_D_L
```

```
## [1] 14.21464
```

```
critic_fr2<-qnorm(p = ratio, mean = mean_D_L, sd = sigma_D_L)
ceiling(critic_fr2)
```

```
## [1] 361
```

```
cbind(ceiling(critic_fr1), ceiling(critic_fr2))
```

```
##      [,1] [,2]
## [1,]  351  361
```

The optimal quantity **(q)** is higher with the bootstrap method and is equal to 361 units.

Another comment can be made about the difference in the estimation of the quantity if the products are perishable or not :

```
cbind(ceiling(mean(D_L)), ceiling(critic_fr2))
```

```
##      [,1] [,2]
## [1,]  351  361
```

The critical fractile is higher than the quantity we expected value from 5000 simulations of demands.

How is this result influenced by the price and the cost ?

Let's keep the price fixed, and we increase the cost to 90.

```
ratio2<-((120-90)/120)
critic_fr3<-qnorm(p = ratio2, mean = mean_D_L, sd = sigma_D_L)
ceiling(critic_fr3)
```

```
## [1] 342
```

Now the quantity from the Newsvendor Model is lower than the one from simulations.

Indeed, that's what the theory behind was showing. The first ratio being larger *(the profit for one more quantity is probably higher in the case where p=120 and c=30 even under uncertainty)* than in the second situation where p=120 and c=90, thus the ordered quantity for the stock can be larger until we satisfy the condition.

List of references :

- Mithu, A. H. and Roy, D. (2013). Identification and reduction of the major factors of demand uncertainty on supply chain network, *Journal of Business and Management, Volume 12, Issue 6, PP 70-77.*
- Pittavino, M. (2019). Lecture 5: PI and distributions h-steps ahead, simulations & Lecture 6: Follow-up on Simulations and Bootstrap, *University of Geneva.*
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